

CoM-VLA: Coordinated Multi-Robot Vision-Language-Action Model for Decentralized Formation Reconfiguration

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Abstract— Coordinated navigation is a fundamental capability for robot teams operating in complex environments. To achieve effective team behavior, robots often need to maintain and adapt different formations across diverse environments. Existing approaches often rely on prior environmental information, limiting their adaptability during formation reconfiguration. Vision-Language-Action (VLA) models unify visual perception, reasoning, and action generation, but remain primarily focused on single-robot navigation, leaving decentralized multi-robot navigation with adaptive formation control largely unexplored. To address these challenges, we introduce Coordinated Multi-Robot VLA (CoM-VLA), an approach for decentralized navigation with adaptive formation reconfiguration. CoM-VLA grounds formation instructions in visually inferred environmental constraints and feasible formations, and learns instruction-grounded adaptive communication to construct team-aware representations. Each robot then independently executes a shared VLA policy conditioned on its local information and team-aware representation. CoM-VLA is initialized from a pretrained single-robot navigation backbone and trained end-to-end through imitation learning. Experiments in simulation and on physical robots demonstrate that CoM-VLA enables decentralized multi-robot navigation with adaptive formation reconfiguration while maintaining high navigation success and formation accuracy.

I. INTRODUCTION

Multi-robot systems have emerged as a promising paradigm for addressing complex tasks by leveraging teams of robots operating within shared environments, with real-world applications in search and rescue [1]–[3], space exploration [4], [5], and warehouse logistics [6]–[8]. A fundamental capability underlying many of these applications is coordinated multi-robot navigation, where robots must navigate complex environments while maintaining effective team behaviors. To achieve this, robot teams often adopt structured formations that improve task performance and operational efficiency.

Different formations, however, are better suited to different situations. For example, wedge formations can provide broad sensing coverage, whereas column formations are more effective for traversing narrow or constrained spaces. As a result, no single formation is universally optimal. To maintain effective coordination across changing environments, robot teams must perform *formation reconfiguration*, adaptively adjusting their spatial configurations in response to environmental constraints. These reconfigurations also require *adaptive coordination*, as changes in formation alter the coordination dependencies among robots, requiring each robot to remain aware of relevant teammates across different formations, together with *decentralized control* to make decisions based on local observations.

Learning-free formation control commonly combines leader-follower strategies [9]–[12] with potential-field [13]–[15] or

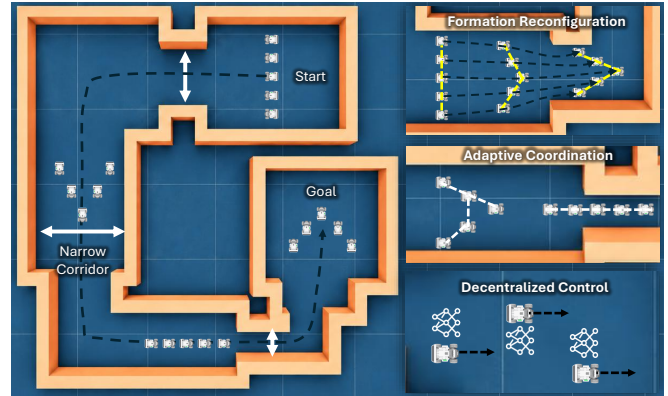


Fig. 1. As the robot team navigates through environments with varying corridor widths, robots must reconfigure their formations, adaptively coordinate with relevant teammates across different formations, and generate decentralized actions through a shared policy to achieve coordinated navigation.

optimization-based control [16]–[18]. Learning-based methods enable adaptive and decentralized coordination through reinforcement learning [19]–[21] and graph learning [22]–[25], while large language models plan formation through scene reasoning [26], [27]. However, these methods often rely on prior environmental information, limiting online formation control. More recently, Vision-Language-Action (VLA) models have advanced single-robot navigation by unifying perception, reasoning, and action generation with reduced reliance on environmental representations [28]–[31]. However, existing VLAs are primarily learned from single-robot demonstrations, grounding actions in individual observations without explicit awareness of teammates. This limits their ability to capture evolving inter-robot dependencies for coordination during formation reconfiguration [32]–[34]. A direct extension to multi-robot coordination is to aggregate team observations within a centralized VLA policy. However, the input context and inference cost grow with team size, limiting scalability to larger robot teams.

We propose **Coordinated Multi-robot VLA (CoM-VLA)**, an approach for decentralized multi-robot navigation with adaptive formation reconfiguration. It analyzes onboard visual observations to estimate the navigable width and applies geometric feasibility rules to identify feasible formation candidates. The final formation is specified by a language instruction or, when no external instruction is provided, automatically selected from the feasible candidates. It then combines the formation instruction with each robot’s local state to infer an instruction-conditioned dynamic communication graph,

where nodes represent robots and edges determine selective information exchange among physically connected teammates. Graph message passing aggregates the received information into a team-aware representation for each robot. Conditioned on its local image, formation instruction, and team-aware representation, each robot independently executes a shared VLA policy to generate its action. CoM-VLA is initialized from a pretrained single-robot navigation VLA and trained end-to-end through imitation learning on demonstrations generated by a centralized expert, jointly optimizing the graph-based coordination module and the VLA policy.

Our main contribution is CoM-VLA, one of the first multi-robot VLA approaches for navigation with decentralized formation reconfiguration. It introduces two key novelties:

- We propose a novel decentralized multi-robot VLA that extends VLA from individual action generation to team-level coordination, enabling robot teams to visually identify environmentally feasible formations, selectively coordinate with relevant teammates to realize formation instructions through adaptive communication, and generate decentralized actions through a shared VLA policy executed independently by each robot.
- We introduce a novel instruction-grounded team coordination method, which jointly embeds formation instructions and local robot states to predict pairwise communication probabilities and dynamically construct the communication topology. This addresses the time-varying coordination requirements during formation reconfiguration, enabling sparse and temporally consistent information exchange for decentralized VLA action generation.

II. RELATED WORK

A. Multi-Robot Navigation with Formation Coordination

We categorize multi-robot navigation with formation coordination into non-learning-based and learning-based approaches. Non-learning-based methods commonly adopt leader-follower coordination, where one or more robots guide the team and followers regulate their relative states to maintain a prescribed formation [9]–[12]. Under such coordination structures, robot motions can be generated using potential-field methods [13]–[15], which encode formation maintenance and obstacle avoidance through attractive and repulsive interactions, or optimization-based methods [16]–[18], which compute coordinated motions by optimizing formation objectives subject to collision-avoidance and motion constraints.

Recently, learning-based methods have been developed to enable more adaptive formation coordination. Reinforcement learning approaches [19]–[21] learn control policies through interaction with the environment, enabling robots to adapt their motions and formations to varying spatial constraints. Graph-learning methods [23]–[25] represent robot teams as graphs and learn inter-robot information aggregation to capture coordination dependencies among teammates, supporting scalable and decentralized decision-making. Large language models have also been explored for formation planning [26], [27], leveraging semantic scene understanding and high-level reasoning to maintain or decompose team formations

according to environmental requirements. However, these methods often rely on prior environmental information, such as maps or predefined formation plans, limiting their ability to perform online formation reconfiguration in response to changing environmental constraints.

B. Vision-Language-Action Models for Robotics

Vision-Language-Action (VLA) models have emerged as a unified framework for integrating visual perception, language understanding, and action generation in robotics. Early VLA models, including RT-2 [35], OpenVLA [36], and π_0 [37], leverage pretrained vision-language representations and large-scale robot demonstrations to generalize robot control across tasks and embodiments.

Recent VLA methods have been extended to single-robot navigation with different designs. NaVILA [38] uses hierarchical control to convert visual-language reasoning into intermediate motion commands executed by a low-level locomotion policy, while OmniVLA [28] conditions navigation on multiple goal modalities to improve task generalization. MM-Nav [30] incorporates multi-view visual observations and learns navigation behaviors from multiple expert policies. AsyncVLA [29] improves control frequency by decoupling high-level semantic reasoning from reactive control, whereas TIC-VLA [31] explicitly accounts for reasoning latency during action generation in dynamic environments.

Despite these advances, existing VLA navigation approaches predominantly focus on single-robot settings, generating actions from individual observations without explicit awareness of teammates, which limits their ability to capture the inter-robot coordination required for multi-robot navigation. A straightforward extension is to centralize team observations within a single VLA policy; however, this causes the input context and inference cost to scale with team size. More recently, CHORUS [39] extends VLA policy to multi-robot collaboration by enabling each robot to infer teammate behavior from local visual observations. However, it relies on teammates being visually observable to the controlled robot, which limits deployment in larger robot teams where maintaining visibility of all teammates becomes increasingly difficult.

III. APPROACH

A. Problem Formulation

We represent a team of n robots as a dynamic directed graph $\mathcal{G} = (\mathcal{V}, \mathbf{A})$, where $\mathcal{V} = \{1, \dots, n\}$ denotes the robots and $\mathbf{A} \in \{0, 1\}^{n \times n}$ is the adjacency matrix that encodes the communication topology. Each entry A_{ij} indicates whether robot i receives information from robot j , where $A_{ij} = 1$ denotes an active communication edge and $A_{ij} = 0$ otherwise. Each robot $i \in \mathcal{V}$ has a local state $\mathbf{s}_i = (\mathbf{g}_i, \mathbf{p}_i)$, where $\mathbf{g}_i \in \mathbb{R}^2$ denotes the relative position of the team goal with respect to the robot, and $\mathbf{p}_i \in \mathbb{R}^2$ denotes the relative position of its assigned formation slot. Each robot also receives a visual observation $\mathbf{I}_i$ and a language instruction τ_i specifying the robot to be controlled and the desired team formation. We further define the action of robot i as $\mathbf{a}_i \in \mathbb{R}^2$, representing the commanded velocities along the x - and y -axes.

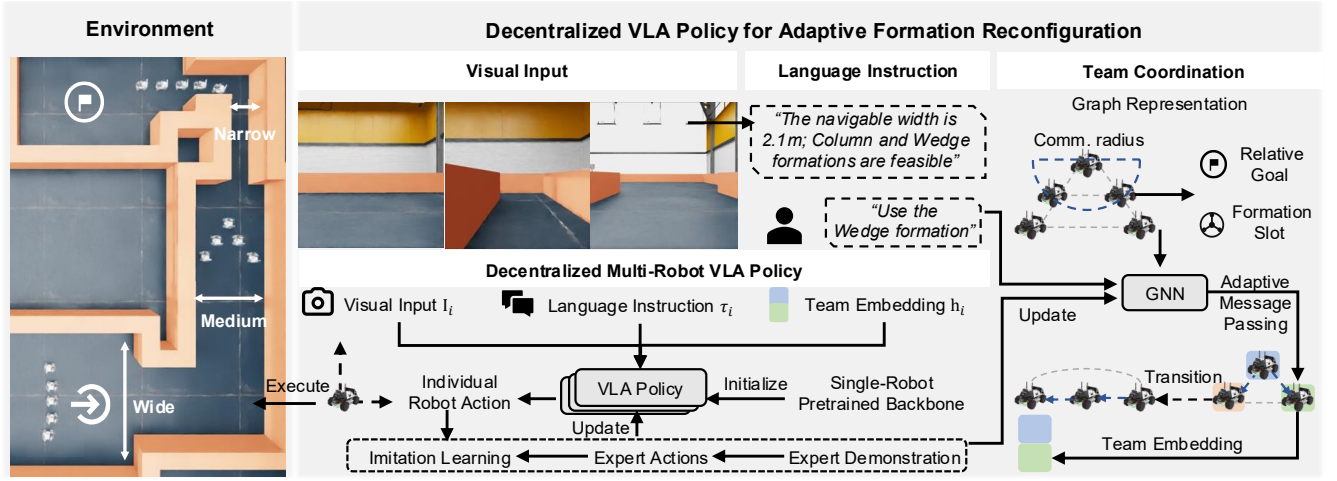


Fig. 2. CoM-VLA grounds formation instructions in visually inferred environmental constraints and feasible formations, and learns instruction-grounded communication to construct team-aware representations. Each robot then independently executes a shared VLA policy conditioned on its local information and team-aware representation. CoM-VLA is initialized from a pretrained single-robot navigation backbone and trained end-to-end through imitation learning.

Given a pretrained single-robot navigation VLA policy π_θ parameterized by θ , a centralized multi-robot VLA policy can be formulated as

$$\{\mathbf{a}_1, \dots, \mathbf{a}_n\} = \pi_\theta(\{\mathbf{I}_i\}_{i=1}^n, \{\tau_i\}_{i=1}^n, \{\mathbf{s}_i\}_{i=1}^n), \quad (1)$$

where $\{\mathbf{I}_i\}_{i=1}^n, \{\tau_i\}_{i=1}^n, \{\mathbf{s}_i\}_{i=1}^n$ denote the visual observations, language instructions, and states of all robots in the team, respectively.

However, multi-robot formation control introduces three unique challenges. First, unlike single-robot commands, formation instructions can be under-specified. For example, a human operator may not know whether a desired formation can safely traverse a narrow passage. The robot team must therefore perceive environmental constraints and report feasible formations for human selection. Second, the communication topology of the robot team is time-varying due to changes in relative distances, e.g., during formation reconfiguration. Robots must therefore coordinate under a shared formation instruction while adapting their communication to changes in team configuration. Third, the VLA policy must support decentralized execution, allowing each robot to independently generate actions from locally available information and enabling scalable deployment to larger robot teams.

B. Vision-Grounded Multi-Robot Formation Instruction

To address the first challenge, CoM-VLA introduces a vision-grounded formation instruction generation method that analyzes each robot’s visual observation and utilizes rule-based prompts to report environmental constraints and feasible formation candidates.

Specifically, given the visual observation $\mathbf{I}_i$ from the i -th robot, CoM-VLA first estimates the navigable width of the upcoming path. The feasible formation candidates are then selected as follows:

$$\mathcal{F}_i^K = \text{TopK}(\{f \in \mathcal{F} \mid s_f + 2\delta \leq w(\mathbf{I}_i)\}, s_f), \quad (2)$$

where $\mathcal{F} = \{f_i\}$ denotes the set of all available formations, s_f denotes the lateral space required by each formation f , $w(\mathbf{I}_i)$ denotes the navigable width along the goal direction estimated from the monocular observation using known camera geometry, and δ is a safety margin on each side of the formation. $\text{TopK}(\cdot, s_f)$ returns the top- K feasible formations $\mathcal{F}_i^K$ with the largest s_f , which are then incorporated into a prompt and reported to the operator, e.g., The navigable width is 2.1 m; Column and Wedge formations are feasible.

Based on this information and the mission objective, the operator selects a desired formation through a language command, e.g., Use the Wedge formation. The command is then converted into the formation instruction τ . To notice, the operator command can be provided once, multiple times, or never during navigation. If no operator command is received, CoM-VLA automatically selects the top-1 feasible formation as the final formation instruction.

C. Instruction-Grounded Team Coordination

During formation reconfiguration, inter-robot coordination requirements evolve with the team configuration as relative distances change. Robots must therefore coordinate under a shared formation instruction while adapting their communication accordingly. To address this, we design a graph neural network ψ that embeds the formation instruction τ with each robot’s local state to predict pairwise communication probabilities p_{ij} . These probabilities dynamically determine the communication topology $\mathbf{A}$, enabling selective information exchange to capture evolving inter-robot dependencies during formation reconfiguration. Given $\mathbf{A}$, ψ performs message passing to generate team-aware embeddings $\{\mathbf{h}_i\}_{i=1}^n = \psi(\mathcal{G})$ for multi-robot VLA action generation.

Specifically, for each robot i , the network ψ first projects its local state $\mathbf{s}_i$ and formation instruction τ_i into a latent representation $\mathbf{z}_i = \mathbf{W}^z[\mathbf{s}_i \parallel \tau_i]$, which captures information from both the language instruction and the robot state. Each

robot maintains its own latent representation under the same instruction. The team-level latent representation is denoted as $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_n]^\top$. $\mathbf{W}^z$ is a learnable weight matrix shared across all robots. For each robot pair, let $\mathbf{d}_{ij} \in \mathbb{R}^2$ denote the relative position of robot j expressed in robot i 's body frame. For pairs within the communication radius r , i.e., $\|\mathbf{d}_{ij}\|_2 \leq r$, the communication probability from robot j to robot i is computed as

$$p_{ij} = \sigma(\mathbf{W}^p[\mathbf{z}_i \parallel \mathbf{d}_{ij}]), \quad (3)$$

where $\mathbf{W}^p$ is shared across all robot pairs and $\sigma(\cdot)$ denotes the sigmoid function. For $\|\mathbf{d}_{ij}\|_2 > r$, we set $p_{ij} = 0$.

Given the communication probabilities $\mathbf{P} = [p_{ij}]_{i,j=1}^n$, we model each communication edge A_{ij} as a Bernoulli variable parameterized by p_{ij} , yielding the topology distribution

$$p(\mathbf{A} \mid \mathbf{P}) = \prod_{i=1}^n \prod_{j \neq i} p_{ij}^{A_{ij}} (1 - p_{ij})^{1-A_{ij}}. \quad (4)$$

Self-edges are excluded, and the directed edges A_{ij} and A_{ji} are determined independently. During training, we employ a straight-through Gumbel-Sigmoid estimator [40] to enable differentiable sampling from $p(\mathbf{A} \mid \mathbf{P})$. During execution, $\mathbf{A}$ is obtained as $A_{ij} = \mathbb{I}[p_{ij} > \delta]$, where $\mathbb{I}[\cdot]$ denotes the indicator function and δ is the communication probability threshold.

After $\mathbf{A}$ is determined, robots selectively exchange information only through active communication edges. We then perform graph message passing, where each robot i aggregates the received teammate representations with its local representation to obtain the team-aware embedding:

$$\mathbf{h}_i = \mathbf{W}^s \mathbf{z}_i + \sum_{j \neq i} A_{ij} \mathbf{W}^m \mathbf{z}_j, \quad (5)$$

where $\mathbf{W}^s$ and $\mathbf{W}^m$ are learnable weight matrices for self and teammate representations, respectively.

To learn adaptive team communication, we regularize both edge density and temporal variation. We minimize the expected fraction of active edges using $\frac{\lambda_c}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i} p_{ij}^t$ to suppress unnecessary message exchange, where λ_c controls the strength of communication sparsity regularization and t denotes the current control timestep. To encourage stable edge connectivity during formation maintenance while allowing the topology to adapt during formation reconfiguration, we additionally minimize $\frac{\lambda_t}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i} |p_{ij}^t - p_{ij}^{t-1}|$, where λ_t controls the temporal consistency of the topology. Formally, we define the communication objective as

$$\mathcal{L}_{\text{comm}} = \underbrace{\frac{\lambda_c}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i} p_{ij}^t}_{\text{Communication sparsity}} + \underbrace{\frac{\lambda_t}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i} |p_{ij}^t - p_{ij}^{t-1}|}_{\text{Temporal consistency}}, \quad (6)$$

which jointly promotes efficient team coordination and temporal stability of the learned topology.

D. Decentralized Multi-Robot VLA Policy

The multi-robot VLA policy must support decentralized execution, allowing each robot to independently generate actions from locally available information and enabling scalable deployment to larger robot teams.

Through graph message passing, the team-aware embedding $\mathbf{h}_i$ obtained in Eq. 5 aggregates the robot's local representation with information from neighboring robots according to the learned communication topology. Given this, each robot generates its action as

$$\mathbf{a}_i = \pi_\theta(\mathbf{I}_i, \tau_i, \mathbf{h}_i), \quad (7)$$

where π_θ denotes the shared decentralized multi-robot VLA policy deployed on each robot. During execution, each robot independently executes π_θ to generate its action from its local visual observation, team-aware representation, and formation instruction, where the instruction is either provided by a human operator or automatically selected from the visually identified feasible formation candidates.

To train CoM-VLA, we initialize it from a pretrained single-robot navigation backbone that provides strong priors for individual navigation and obstacle avoidance, and train the entire approach end-to-end through imitation learning, jointly optimizing the navigation policy and graph neural network. The expert demonstrations are generated by a *centralized* model predictive control (MPC) planner that uses *global* robot states and environmental information to optimize expert actions $\mathbf{a}_i^*$ to achieve the desired formation and navigation objectives. For each robot i , CoM-VLA takes its visual observation $\mathbf{I}_i$, language instruction τ_i , and state $\mathbf{s}_i$ as input. The graph network ψ first constructs the team communication topology $\mathbf{A}$ and produces a team-aware embedding $\mathbf{h}_i$ for each robot through message passing. The shared VLA policy π_θ predicts the action distribution conditioned on $\mathbf{I}_i$, τ_i , and $\mathbf{h}_i$. Given the expert action $\mathbf{a}_i^*$, we first discretize it into action tokens and optimize the policy using the cross-entropy objective:

$$\mathcal{L}_{\text{policy}} = - \sum_{i=1}^n \log \pi_\theta(\mathbf{a}_i^* \mid \mathbf{I}_i, \tau_i, \mathbf{h}_i), \quad (8)$$

where the objective is minimized using the Adam optimizer [41]. The gradients of $\mathcal{L}_{\text{comm}}$ in Eq. 6 are backpropagated through ψ , while the gradients of $\mathcal{L}_{\text{policy}}$ in Eq. 8 are backpropagated through both ψ and π_θ , enabling joint optimization of the communication topology and action policy. For π_θ , we adopt Low-Rank Adaptation (LoRA) [42] to adapt a pretrained single-robot navigation VLA backbone to coordinated formation reconfiguration by updating only the low-rank adapters while keeping the backbone frozen.

IV. EXPERIMENTS

A. Experimental Setups

We evaluate CoM-VLA in two settings: (1) simulation in NVIDIA Isaac Sim and (2) physical multi-robot experiments using ROS 2. Across both settings, we construct maze-like environments with diverse corridor configurations, including narrow passages, long corridors, and turning corridors. These scenarios require the robot team to perceive the environment, identify environmental constraints, and determine formation candidates. The robots maintain the desired formation when feasible and adaptively reconfigure among line, wedge, and

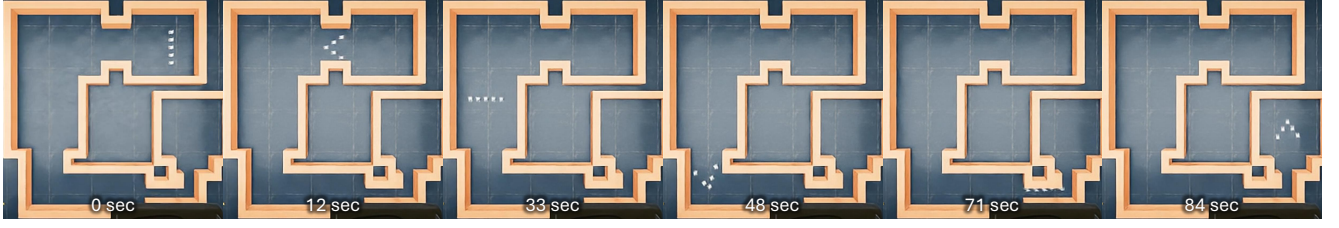


Fig. 3. Qualitative results of coordinated navigation with adaptive formation reconfiguration using CoM-VLA in Isaac simulation. Each robot independently executes a shared VLA policy to navigate a long maze environment with diverse corridor geometries, requiring the team to adapt among different formation types to safely and efficiently reach the goal.

TABLE I

QUANTITATIVE COMPARISON OF CoM-VLA AND BASELINE METHODS ACROSS DIFFERENT FORMATION RECONFIGURATION SCENARIOS IN SIMULATION.

Method	Line → Wedge				Wedge → Column				Column → Line			
	SR (%) ↑	FE (m) ↓	RE (m) ↓	TT (s) ↓	SR (%) ↑	FE (m) ↓	RE (m) ↓	TT (s) ↓	SR (%) ↑	FE (m) ↓	RE (m) ↓	TT (s) ↓
DWA [43]	100 ± 0	0.21 ± 0.02	0.11 ± 0.02	12.82 ± 0.21	100 ± 0	0.33 ± 0.08	0.26 ± 0.13	17.82 ± 0.94	100 ± 0	1.29 ± 0.17	1.35 ± 0.10	17.95 ± 1.12
Deform [16]	100 ± 0	0.27 ± 0.02	0.15 ± 0.02	19.18 ± 0.43	100 ± 0	0.29 ± 0.11	0.26 ± 0.07	30.10 ± 0.56	100 ± 0	1.35 ± 0.08	1.38 ± 0.02	27.41 ± 0.96
APF [44]	100 ± 0	0.33 ± 0.03	0.26 ± 0.04	13.77 ± 1.58	100 ± 0	0.32 ± 0.01	0.27 ± 0.01	17.98 ± 0.69	90 ± 4	1.06 ± 0.03	0.78 ± 0.44	19.80 ± 0.60
VLA-LF	100 ± 0	0.22 ± 0.03	0.12 ± 0.01	12.90 ± 0.38	100 ± 0	0.28 ± 0.03	0.25 ± 0.02	17.25 ± 0.72	80 ± 8	0.87 ± 0.02	1.00 ± 0.17	17.32 ± 0.19
CoM-VLA (Ours)	100 ± 0	0.15 ± 0.02	0.04 ± 0.01	12.74 ± 0.26	100 ± 0	0.27 ± 0.03	0.11 ± 0.04	17.88 ± 0.81	100 ± 0	0.84 ± 0.15	0.75 ± 0.31	16.89 ± 0.53

Results are reported as mean ± standard deviation over five seeds. ↑ and ↓ indicate higher and lower values are better, respectively. Best results are shown in bold.

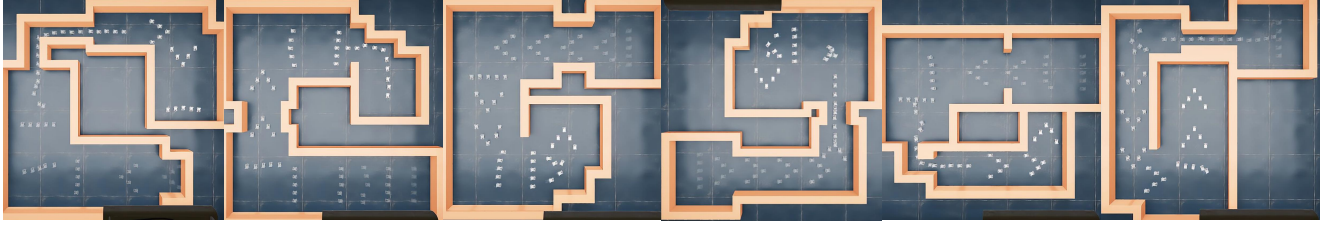


Fig. 4. Qualitative generalization results of CoM-VLA in six representative unseen maze environments. Each subfigure presents a novel maze layout with different corridor geometries and formation-reconfiguration requirements. Representative robot formations are highlighted at different timesteps.

column formations to navigate to the goal. In simulation, robot poses are obtained from the Isaac environment and used to compute relative goal positions, while in physical experiments, robot poses are provided by a motion-capture system.

We build CoM-VLA upon the pretrained OmniVLA navigation backbone [28], which employs fused DINOv2 [45] and SigLIP [46] visual encoders together with a Llama-2 7B language model [47]. We fine-tune the backbone using LoRA with rank 32 and optimize the model with a learning rate of 5×10^{-5} . Training is performed on a single NVIDIA H100 GPU with 80 GB of memory. Additional implementation details are provided in the supplementary material.

We compare CoM-VLA against three motion planners and one VLA baseline: (1) *Dynamic Window Approach (DWA)* [43], which searches over dynamically reachable velocity commands based on goal progress and obstacle clearance; (2) *Deform* [16], which uses model predictive control (MPC) to optimize a short-horizon control sequence by minimizing navigation and formation costs; (3) *Artificial Potential Field (APF)* [44], which uses reactive gradient-based control by combining attraction toward the formation with repulsion from local obstacles and teammates; and (4) *VLA Leader-Follower (VLA-LF)*, a baseline in which the leader robot uses a VLA model for navigation, while the followers maintain their relative positions to the leader. For a fair comparison, all evaluated baselines are restricted to the same local map

observations available to CoM-VLA.

We quantitatively evaluate all methods using four metrics: (1) *Success Rate (SR)*, the percentage of scenarios in which the robot team successfully completes the reconfiguration and reaches the goal; (2) *Formation Error (FE)*, the mean deviation of robots from their assigned formation positions, measured in meters; (3) *Reconfiguration Error (RE)*, the minimum formation error achieved after each formation reconfiguration, measured in meters; and (4) *Travel Time (TT)*, the time required for the team to reach the goal, measured in seconds.

B. Results Using High-Fidelity Robotics Simulation

We first evaluate CoM-VLA in a high-fidelity NVIDIA Isaac Sim environment with five holonomic LIMO robots, each equipped with an onboard RGB camera and executing the shared policy in a decentralized manner. The qualitative results are shown in Figure 3. The environment replicates a maze-like scenario with diverse corridor widths, requiring the robot team to perform long-range navigation while adaptively reconfiguring its formation to safely reach the goal. As the available free space changes along the route, CoM-VLA enables the team to reason about the local environment, choose an appropriate formation, and efficiently reconfigure between different formation types. During each formation stage, the robots tightly maintain the desired formation while performing collision-free navigation. These results demonstrate that CoM-VLA enables the team to adapt its formation throughout a long

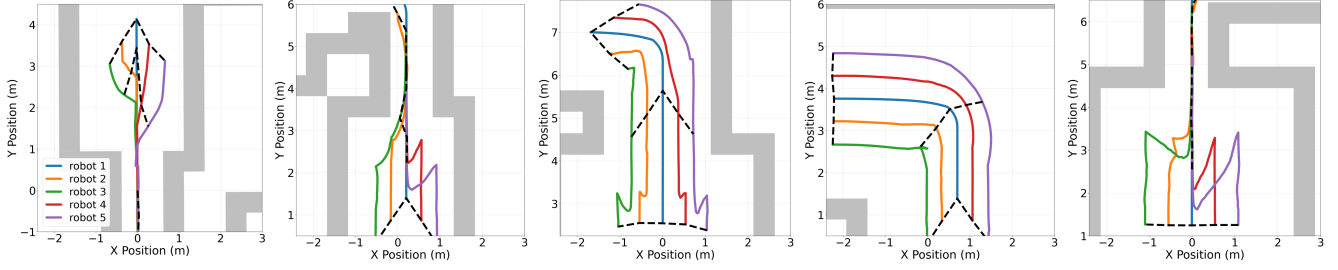


Fig. 5. Representative trajectories of five robots navigating a diverse corridor that requires adaptive formation reconfiguration among different formation types. Each robot’s trajectory is shown in a distinct color, while gray regions indicate obstacles. Black dashed lines illustrate the team formation at selected timestamps.

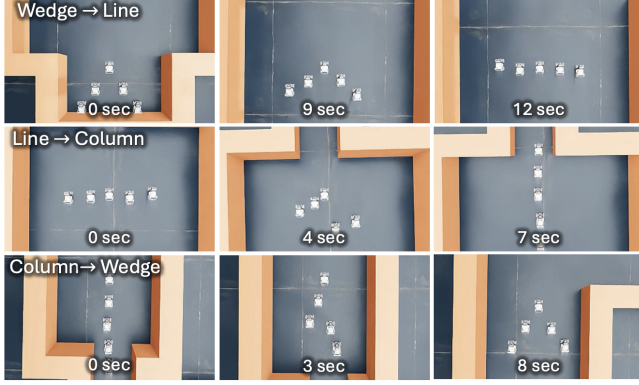


Fig. 6. Qualitative examples of selected formation reconfiguration. Each row shows a different reconfiguration at three representative timestamps, illustrating the team formation before, during, and after the reconfiguration.

navigation task while preserving coordinated motion.

The quantitative results are shown in Table I. DWA achieves short travel times, but its formation performance degrades in more challenging reconfigurations. MPC also maintains a 100% success rate and stable formation control, but requires longer travel times due to its conservative finite-horizon optimization. APF performs well in the first two reconfigurations, but its success rate drops to 90% in the third, with increased formation error. VLA-LF achieves efficient reconfiguration in simpler cases, but its leader-follower structure limits adaptability, resulting in only an 80% success rate in the third reconfiguration. Notably, all baselines are restricted to the same local environmental observations as CoM-VLA, which limits classical planners that typically benefit from broader map information or longer-horizon planning. In contrast, CoM-VLA achieves a 100% success rate across all reconfigurations, consistently obtains the lowest formation error and reconfiguration error, and maintains competitive travel times. Overall, CoM-VLA provides reliable and accurate adaptive formation reconfiguration during decentralized navigation.

C. Results in Unseen Environments

We evaluate the generalization capability of CoM-VLA in unseen environments, as shown in Figure 4. The test set contains multiple maze layouts that are not observed during training, with different corridor geometries, turns, start and goal configurations, and formation-reconfiguration scenarios.

Each subfigure shows one unseen maze, with the robot formations highlighted at representative timesteps along the navigation trajectory. Across these environments, the team consistently adapts its formation shape according to the local corridor geometry and successfully executes the required formation reconfigurations while maintaining coordinated motion toward the goal. These results demonstrate that CoM-VLA generalizes beyond the training layouts and enables coordinated formation reconfiguration in unseen environments.

We visualize the movement trajectories of the robot team using selected examples, as shown in Figure 5. Each colored curve denotes the trajectory of an individual robot, while the gray regions represent obstacles and the black dashed lines indicate the team formation at selected timestamps. The trajectories show that the robots continuously adjust their relative positions as the corridor geometry changes, enabling smooth reconfiguration between different formation types while avoiding obstacles. We further select representative formation-reconfiguration cases, as shown in Figure 6. Each example presents the team configuration before, during, and after the reconfiguration, illustrating how CoM-VLA coordinates the robots to reconfigure and recover the desired formation. These results further demonstrate that CoM-VLA enables robust adaptive formation reconfiguration.

D. Real-World Validation with Physical Robot Teams in ROS2

We validate CoM-VLA through real-world case studies using teams of LIMO robots controlled via ROS2, with selective inter-robot communication implemented through WiFi broadcasting. The experimental environment consists of three corridors with different widths, indicated by different colors, requiring the team to adapt its formation accordingly. The qualitative results are shown in Figure 7. The robot team initially maintains a line formation in the wide-open region to provide broad spatial coverage. As the available space decreases, the team smoothly reconfigures to a wedge formation and maintains this formation while navigating through the narrower passage. Upon perceiving an extremely narrow corridor, the team further reconfigures to a column formation, allowing the robots to safely enter and traverse the constrained region. The team then executes a smooth turn while maintaining coordinated motion and efficiently reaches the goal. These results demonstrate the real-world applicability of CoM-VLA for coordinated multi-robot navigation with adaptive formation

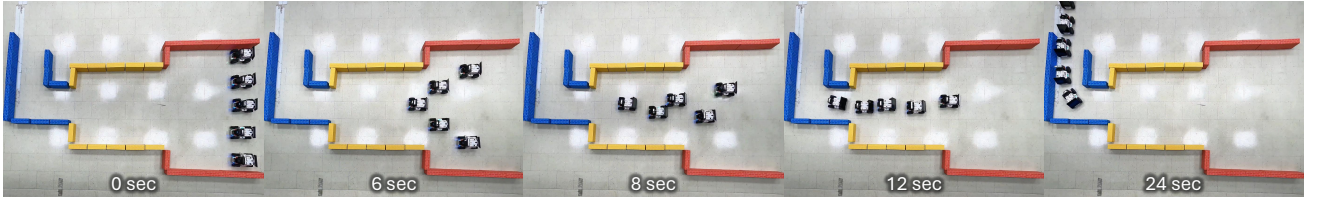


Fig. 7. Real-world validation of CoM-VLA with a physical LIMO robot team. The results show representative stages of navigation through corridor regions with different widths, where the team adapts its formation from line to wedge and then to column while maintaining coordinated motion toward the goal.

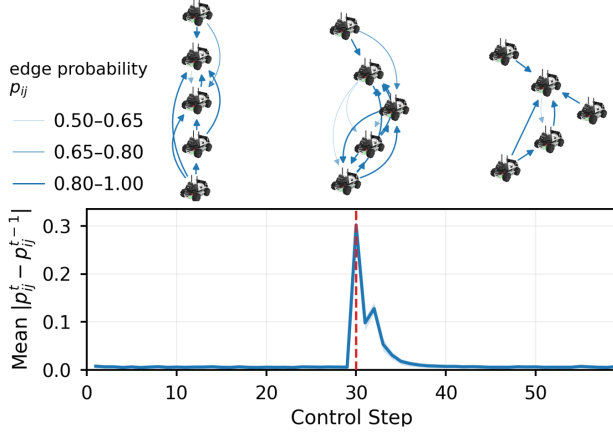


Fig. 8. Learned communication topology during a formation reconfiguration. The top figure shows three stages of the reconfiguration, with edge strength indicating the learned communication probability p_{ij} , while the bottom figure shows the mean temporal variation of p_{ij} across control steps.

reconfiguration in challenging environments.

E. Effects of Instruction-Grounded Team Coordination

We analyze the effectiveness of instruction-grounded team coordination in CoM-VLA through the learned communication probabilities p_{ij} , as shown in Figure 8. The top figure visualizes the learned communication topology before, during, and after a line-to-wedge formation reconfiguration. First, CoM-VLA learns sparse coordination by selectively exchanging information with relevant teammates, avoiding unnecessary fully connected communication. Second, the learned communication topology varies with the formation instruction: line and wedge formations exhibit distinct edge structures, indicating that CoM-VLA captures formation-dependent coordination dependencies. Third, the bottom figure shows the temporal evolution of p_{ij} throughout the reconfiguration. The learned communication topology remains nearly unchanged while the team maintains a stable formation, changes rapidly when reconfiguration is triggered, and gradually converges to a new stable topology afterward. These results further demonstrate that instruction-grounded team coordination enables adaptive, sparse, and temporally stable coordination during formation reconfiguration.

F. Scalability to Larger Robot Teams

We analyze the scalability of CoM-VLA with respect to team size, with qualitative results shown in Figure 9. Using line-to-wedge reconfiguration as a representative example, we evaluate

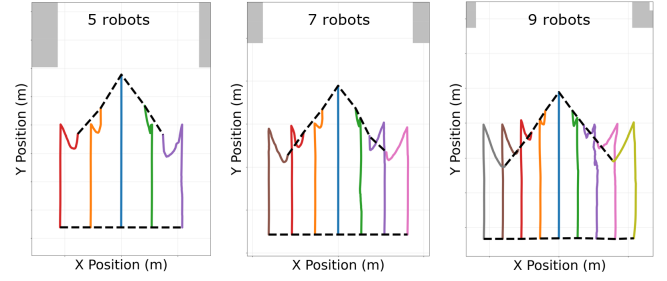


Fig. 9. Qualitative scalability results of CoM-VLA. Representative line-to-wedge reconfiguration trajectories for teams of 5, 7, and 9 robots, demonstrating consistent formation reconfiguration across increasing team sizes.

teams of 5, 7, and 9 robots. Across increasing team sizes, CoM-VLA consistently completes coordinated navigation and formation reconfiguration, with each robot independently executing the shared policy while selectively coordinating with relevant teammates. We further evaluate scalability quantitatively using formation error (FE) and reconfiguration error (RE). CoM-VLA achieves an FE of 0.115 m and a RE of 0.012 m with 5 robots, compared with an FE of 0.154 m and a RE of 0.054 m with 9 robots, showing that coordination performance remains effective as team size increases. Similar behavior is observed across other formation reconfigurations. Together, both qualitative and quantitative results demonstrate that CoM-VLA scales effectively to larger robot teams.

V. CONCLUSION

We introduce CoM-VLA, a multi-robot VLA approach for decentralized navigation with formation reconfiguration. CoM-VLA grounds formation instructions in visually inferred environmental constraints and feasible formations, and learns instruction-grounded adaptive communication to construct team-aware representations. Conditioned on its local information and team-aware representation, each robot independently executes a shared VLA policy for decentralized action generation. CoM-VLA is initialized from a pretrained single-robot navigation backbone and trained end-to-end through imitation learning. Experiments across simulation and real-world environments demonstrate that CoM-VLA enables effective formation reconfiguration, adaptive and sparse team coordination, and scalable decentralized navigation. One limitation is that CoM-VLA uses a predefined formation set. Future work will explore language-conditioned formation generation, where the model directly infers formation geometry from observations, enabling more diverse formation reconfigurations.

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